

CS3001 – Computer Networks

Quiz – 3A

Name	
Roll. No.	
Section	

CLO1: Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack.

Part 1 [7 Marks]:

In the rdt3.0 protocol the sender uses only two sequence numbers $\{0, 1\}$. Assume the physical channel never corrupts or loses frames, but the sender's timeout is **shorter than the one-way propagation time**, so the timer always expires before the very first ACK can arrive.

Starting from the instant the application hands the first data segment to the transport layer, describe:

1. **(In short bullet points, step-by-step)** Events, state transitions, and messages exchanged until the sender returns to state “Wait for call 1 from above”.
2. How many data packets and how many ACKs (both of which type) are actually transmitted on the wire
3. Why does the receiver still deliver the data only once?

Solution:

1.
 - Sender begins in “Wait for call 0 from above”, sends pkt0, starts the timer, and enters “Wait for ACK0”.
 - Because the $\text{timeout} < \text{RTT}$, the **timeout fires first**, so the sender retransmits the identical pkt0, restarts/resets/starts-again the timer, and stays in “Wait for ACK0”.
 - The receiver sees the *first* arrival of pkt0, verifies its checksum, accepts sequence 0 as expected, delivers the payload up once, and replies with ACK0.
 - Shortly afterwards the *duplicate* pkt0 arrives; the receiver recognises the repeated sequence 0, suppresses a second delivery, but still emits another ACK0 (duplicate ACK).
 - The **first ACK0** (or the duplicate, whichever arrives first) reaches the sender, which stops the timer and transitions to “Wait for call 1 from above”.
2. Two data packets (both seq 0) and two ACKs (both ACK0).
3. The receiver's state machine discards duplicates after sequence-number comparison.

Part 2 [3 Marks]:

A Go-Back-N (GBN) sender has window size $N = 4$. The receiver expects sequence number 5 next ($\text{rcv_base} = 5$). Packets arrive at the receiver in this order: 7, 5, 6, 5.

1. Which of these packets does the receiver deliver to the application (and in what order)?

2. Which ACKs does the receiver send (list them in order sent)?
3. After processing all four arrivals, what is the receiver's new expected sequence number (rcv_base)?

Solution:

1. Delivered (in order): 5, 6
2. ACKs sent (in order): ACK4, ACK5, ACK6, ACK6
3. New expected sequence number (rcv_base): 7.